

DECOWA : Deformation-Constrained Warping Attack

Transfer Attacks in Face Recognition

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Attacks Considered

- ☐ PGD (baseline)
- ☐ MI-FGSM (momentum-based)
- ☐ TI-FGSM (translation-invariant)
- ☐ SI-NI-FGSM (scale-invariant Nesterov)
- ☐ MI-ADMIX-DI-TI (admixed input diversity)
- ☐ BPA-CNN (backward propagation adaptation)
- ☐ **DECOWA (deformation-constrained warping attack)**

Selecting DECOWA

Why DECOWA?

- ❑ Recent publication from the adversarial-robustness literature (AAAI 2024)
- ❑ Achieves state-of-the-art transferability via geometric image warping
- ❑ Fundamentally different from gradient-only methods — applies *spatial deformation*
- ❑ Not implemented in the intern baseline

Code Reference — [GitHub Repo](#)

DECOWA (Deformation-Constrained Warping Attack)

Definition & Nature

- White-box transfer-based adversarial attack
- Gradient-based optimization enhanced with geometric warping
- Adapted for black-box / pre-trained CNN face models
- Uses Thin-Plate-Spline (TPS) deformation to augment inputs during gradient computation

Basic Idea

- Optimizes **spatial warping** of the input, not only pixel perturbations
- Generates multiple deformed copies per iteration
- Averages gradients across deformed copies for a more transferable perturbation
- Warping is constrained via a deformation step (ρ) to prevent extreme distortion
- “Input diversity via geometric deformation” avoids overfitting to the surrogate boundary

Working of DECOWA Attack

Algorithm Steps

- ➊ Initialize adversarial image as the original input
- ➋ Define attack objective (impersonation or dodging)
- ➌ **For T iterations:**
 - Initialize gradient accumulator to zero
 - **For K warping samples:** generate a random TPS noise map; descend the attack loss w.r.t. the noise map ($\rho = 0.01$); warp the image; compute and accumulate the gradient through the warped copy
 - Average gradients over all K warps; normalize by mean absolute value
 - Momentum: $g = \text{DECAY} \cdot g + \text{grad}$; $\text{adv} = \text{adv} + \alpha \cdot \text{sign}(g)$
 - Project into the ε -ball; clip to $[-1, 1]$
- ➍ Return the final adversarial image

- Operates in **embedding space** (cosine similarity)
- **Impersonation**: increase cosine similarity between the adversarial face and the target → make the system treat two different people as the same
- **Dodging**: decrease cosine similarity between the adversarial face and its genuine pair → make the system fail to recognize the same person
- Optimizes embedding distance directly through differentiable warping

Thin-Plate-Spline (TPS) Warping

- 3×3 control mesh over the image
- Interior points displaced by a learnable noise map
- Edge points fixed to prevent extreme distortion
- Smooth, differentiable spatial deformations

Deformation-Constrained Update

- Noise map initialized randomly per warp
- One gradient step finds a “hard” warp — one that makes the objective harder
- Forces the gradient to account for geometric variation $\rightarrow$ better transfer

Gradient Averaging Over Warps

- 20 warping samples per iteration (`DECOWA_NUM_WARPING = 20`)
- Each sample is a different deformed view
- Averaging smooths out model-specific features
- Same spirit as DI-FGSM, but geometric deformation instead of random resizing

Momentum Update

- Like MI-FGSM: accumulated gradient momentum
- $g = \text{decay} \cdot g + \text{grad}$
- $\text{adv} = \text{adv} + \alpha \cdot \text{sign}(g)$

```
ALL_ATTACKS = [  
    'PGD', 'MI_FGSM',  
    'TI_FGSM', 'SI_NI_FGSM',  
    'MI_ADMIX_DI_TI',  
    'DECOWA' # <-- added  
]
```

```
if attack_name == "DECOWA":  
    return decowa(...)
```

Parameter	Value
ε	0.062
NUM_ITER	5
DECAY	1.0
DECOWA_MESH	3 (3 × 3 grid)
NUM_WARPING	20
NOISE_SCALE	2.0
RHO (ρ)	0.01

DECOWA in Our Implementation

- 1 Initializes momentum variable g
- 2 Per iteration, performs **20 warping samples**:
random TPS noise map $\rightarrow$
deformation-constrained update ($\rho = 0.01$) $\rightarrow$
TPS warp via bilinear sampling $\rightarrow$ gradient
through the warped image
- 3 Averages gradients across all 20 warps
- 4 Normalizes by mean absolute value
- 5 Momentum: $g = \text{DECAY} * g + \text{normalized_grad}$
- 6 Standard ε -projection via clipping ($\varepsilon = 0.062$)

Full implementation $\rightarrow$ [GitHub Repo](#)

Combined Leaderboard: All Attacks

Rank	Attack Method	Breach Rate (%)	Impact Mean
1	DECOWA	33.13	0.1854
2	SI_NI_FGSM	28.33	0.1697
3	MI_FGSM	25.62	0.1576
4	MI_ADMIX_DI_TI	22.92	0.1434
5	TI_FGSM	20.62	0.1267
6	PGD	16.67	0.0962

Conclusion

DECOWA achieves the highest overall breach rate of 33.13%, beating all vanilla baselines including SI_NI_FGSM (+4.80 pp) and PGD (+16.46 pp). Its strength is **geometric input diversity**: averaging gradients over 20 spatially-warped copies per iteration yields perturbations that transfer across architectures far better than purely gradient-based methods.

Thank You

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Full Implementation → [GitHub Repo](#)